Digital Image Processing

Lecture 7: Color Image Processing

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Readings: Chapter 7
How to read: relaxed, with notes!

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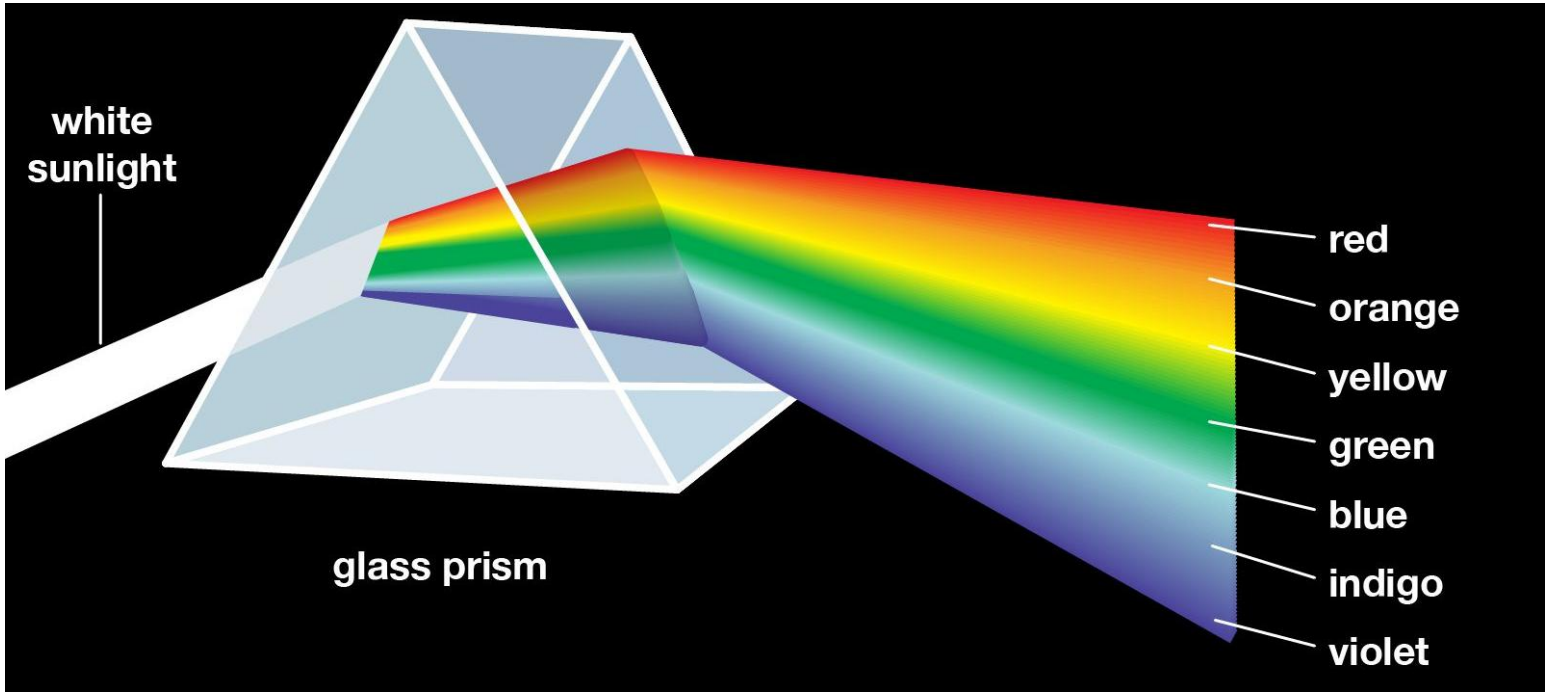
Color Fundamentals

- In 1666, Isaac Newton discovered that when a **beam of sunlight passes through a glass prism**, the emerging light consists of a continuous spectrum of colors, from **violet to red**.
- A body that reflects light that is balanced in all visible wavelength appears white.
- A body that favors reflectance in a limited range of the visible spectrum shows some shades of color.

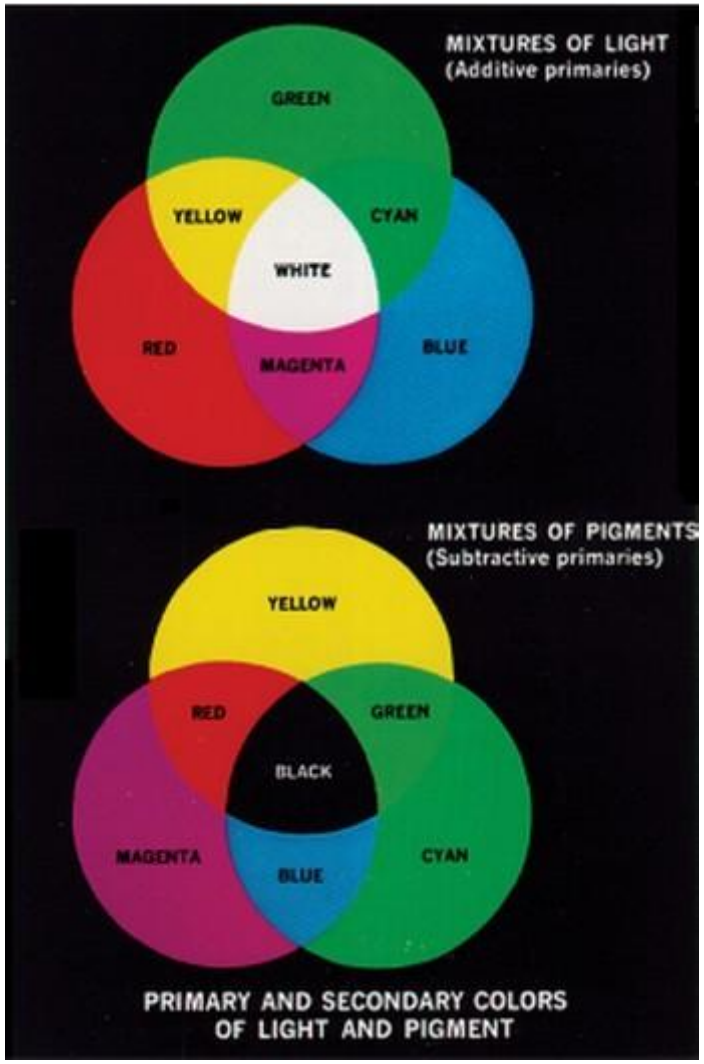


Color Fundamentals

- **Monochromatic** (*achromatic*) light, is a light that is void of color, represented only by its intensity (gray level), ranging from **black to white**.
- **Chromatic** light spans the electromagnetic energy spectrum from **400** to **700** nano-meter.
- **Radiance**: total amount of **energy** that *flows* from the light source, measured in watts (W).
- **Luminance**: amount of energy an observer *perceives* from a light source, measured in lumens (lm).
- **Brightness**: a subjective descriptor of light perception, impossible to measure, representing the achromatic notion of intensity.
- **Cone cells** are responsible for color vision.

Color Fundamentals

- **Standard primary colors**: red (**R**) at 700 nm, green (**G**) at 546.1 nm and blue (**B**) at 435.8 nm.
- Primary does not mean that all visible colors can be generated by primary colors; the wavelength needs to change as well.
- **Secondary colors** can be made by adding primary colors: <u>magenta</u> (**R+B**), <u>cyan</u> (**G+B**) and <u>yellow</u> (**R+G**).



a
b

FIGURE 7.4 Primary and secondary colors of light and pigments. (Courtesy of the General Electric Co., Lighting Division.)

Color Fundamentals

- There is a difference between **primary colors of light** and the ***primary colors of pigments or colorants***.
- In **pigments**, a primary color is the one that subtracts or absorbs a primary color of light, and reflects or transmits the other two.
- The **primary colors of pigments** are **magenta**, **cyan** and **yellow**, and the **secondary colors are red**, **green** and **blue**.
- Distinguishing characteristics of colors:
  - ***Brightness***: achromatic notion of intensity.
  - ***Hue***: dominant wavelength in a mixture of light waves, or the dominant color perceived by an observer.
  - ***Saturation***: relative purity or the amount of white light mixed with the hue.
- **Hue and saturation** combined are called ***chromaticity***.

Color Fundamentals

- The amounts of **red**, **green** and **blue** to form any particular color are called ***tristimulus*** values, and are shown by **X**, **Y** and **Z**.
- The color is specified by its ***trichromatic coefficients*** as:

$$x = \frac{X}{X+Y+Z}$$

$$y = \frac{Y}{X+Y+Z}$$

$$z = \frac{Z}{X+Y+Z}$$

$$x + y + z = 1$$

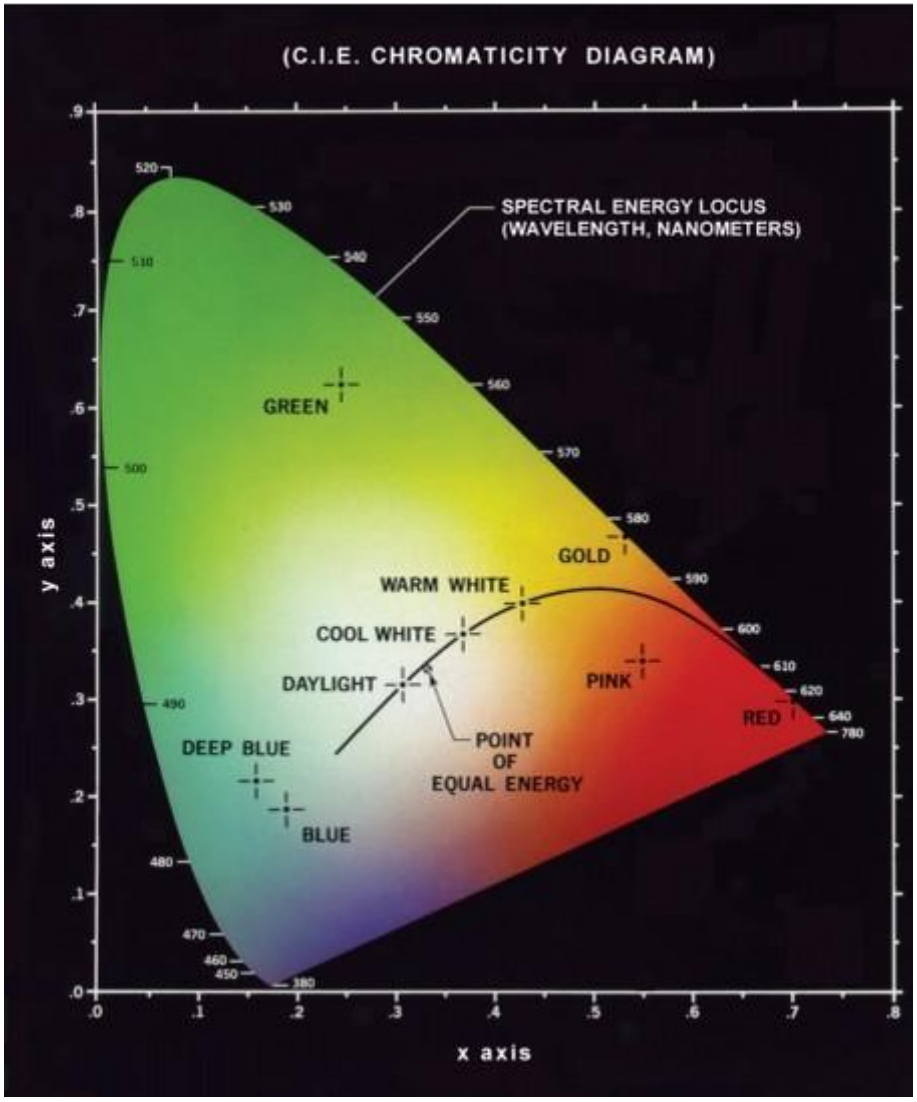


FIGURE 7.5 The CIE chromaticity diagram. (Courtesy of the General Electric Co., Lighting Division.)

Color Models - RGB

- In the **RGB** model, each color appears in its primary spectral components of **red, green and blue**.
- The model is shown in a **Cartesian** coordinate system.
- In the figure, the primary values are at the **three** corners while the secondary colors are at the other three corners.
- **Black** is at the origin, and **white** is at the corner farthest from the origin.
- The **grayscale**, where the RGB values are all **equal**, extends from black to white along the line joining these two points.

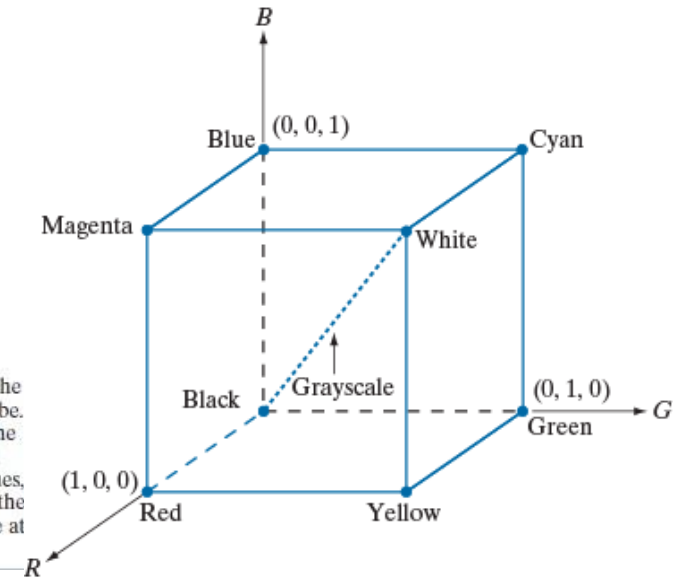


FIGURE 7.7 Schematic of the RGB color cube. Points along the main diagonal have gray values, from black at the origin to white at point (1, 1, 1).

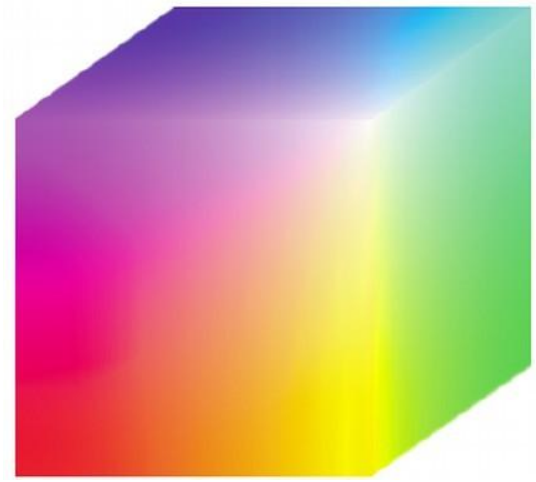


FIGURE 7.8 A 24-bit RGB color cube.

Color Models – CMY and CMYK

- The secondary colors of light (cyan, magenta and yellow) are the primary colors of pigments.
- For example, when a surface coated with cyan pigment is illuminated by white light, no red light is reflected from the surface, since cyan subtracts red light from the white light.
- Conversion from **RGB** to **CMY** is simple:

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

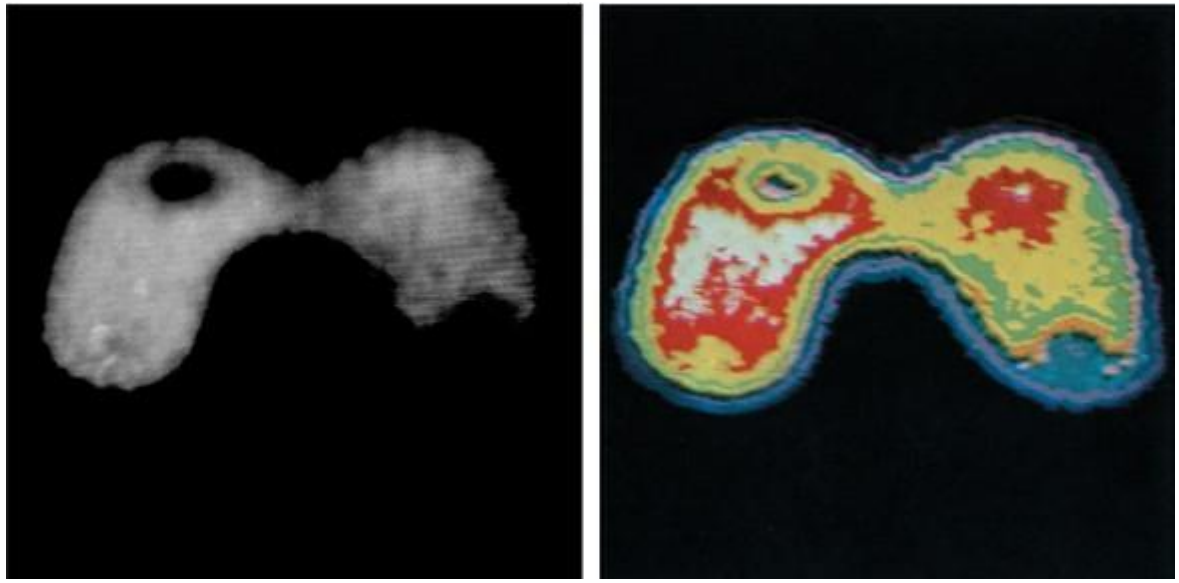
- **Equal amounts** of pigment primaries should produce black.
- In practice, since **C**, **M** and **Y** inks are not pure colors, this cannot be done. Instead, a fourth color black, shown by **K**, is added.

Color Models – HSI

- **RGB** and **CMY** color models are not well-suited for color description for practical human interpretation.
- Humans describe colors by *hue, saturation and brightness*.
- Distinguishing characteristics of colors:
  - ***Brightness***: achromatic notion of intensity.
  - ***Hue***: dominant wavelength in a mixture of light waves, or the dominant color perceived by an observer.
  - ***Saturation***: relative purity or the amount of white light mixed with the hue.
- **Hue** and **saturation** combined are called ***chromaticity***.
- The **HSI** (hue, saturation, intensity) color model, decouples the intensity component from the color-carrying information.

Pseudocolor Image Processing

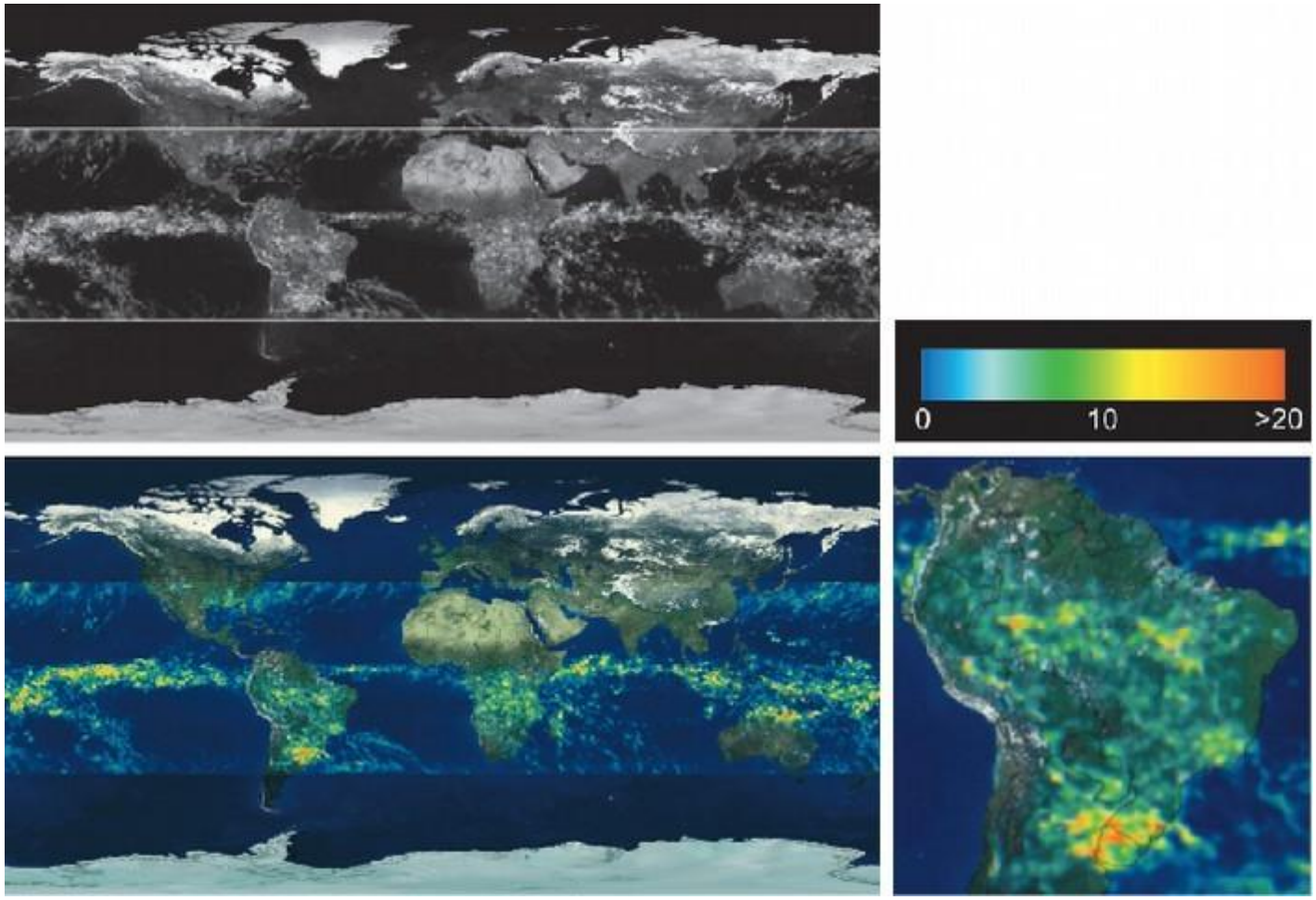
- Pseudocolor image processing consists of ***assigning colors to gray values*** based a specified criterion.
- Humans can distinguish thousands of color shades and intensities compared to less than two ***dozen shades of gray***.
- **Intensity slicing** is the simplest method in pseudocolor image processing.
- In this technique the range from black to white colors are divided into sections, each shown with a different color.



a b

FIGURE 7.18 (a) Grayscale image of the Picker Thyroid Phantom. (b) Result of intensity slicing using eight colors. (Courtesy of Dr. J. L. Blankenship, Oak Ridge National Laboratory.)

Pseudocolor Image Processing

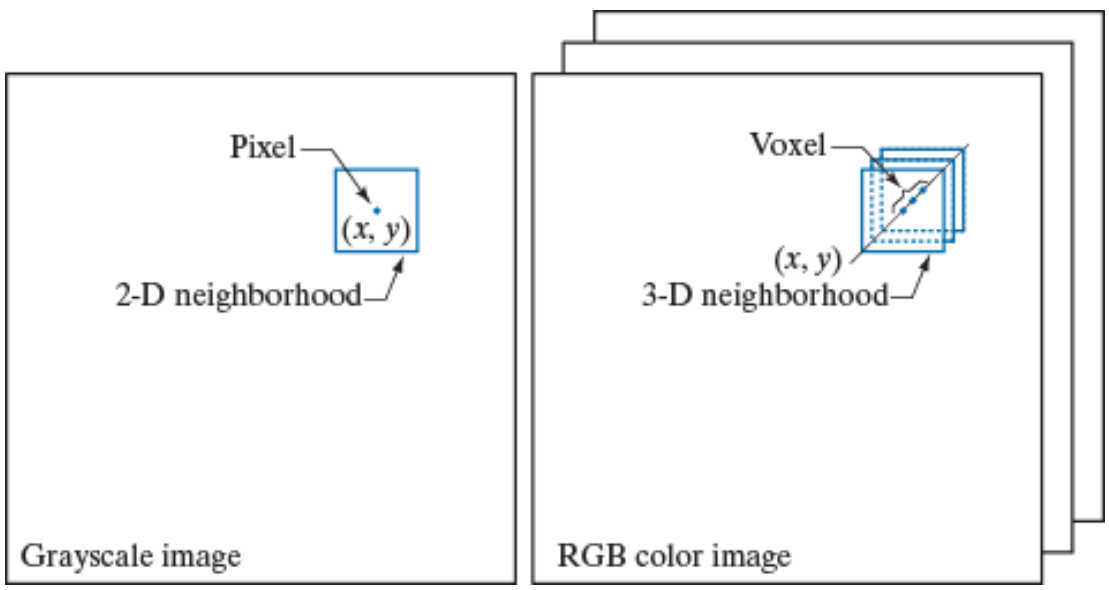


a b
c d

FIGURE 7.20 (a) Grayscale image in which intensity (in the horizontal band shown) corresponds to average monthly rainfall. (b) Colors assigned to intensity values. (c) Color-coded image. (d) Zoom of the South American region. (Courtesy of NASA.)

Basics of Full-Color Image Processing

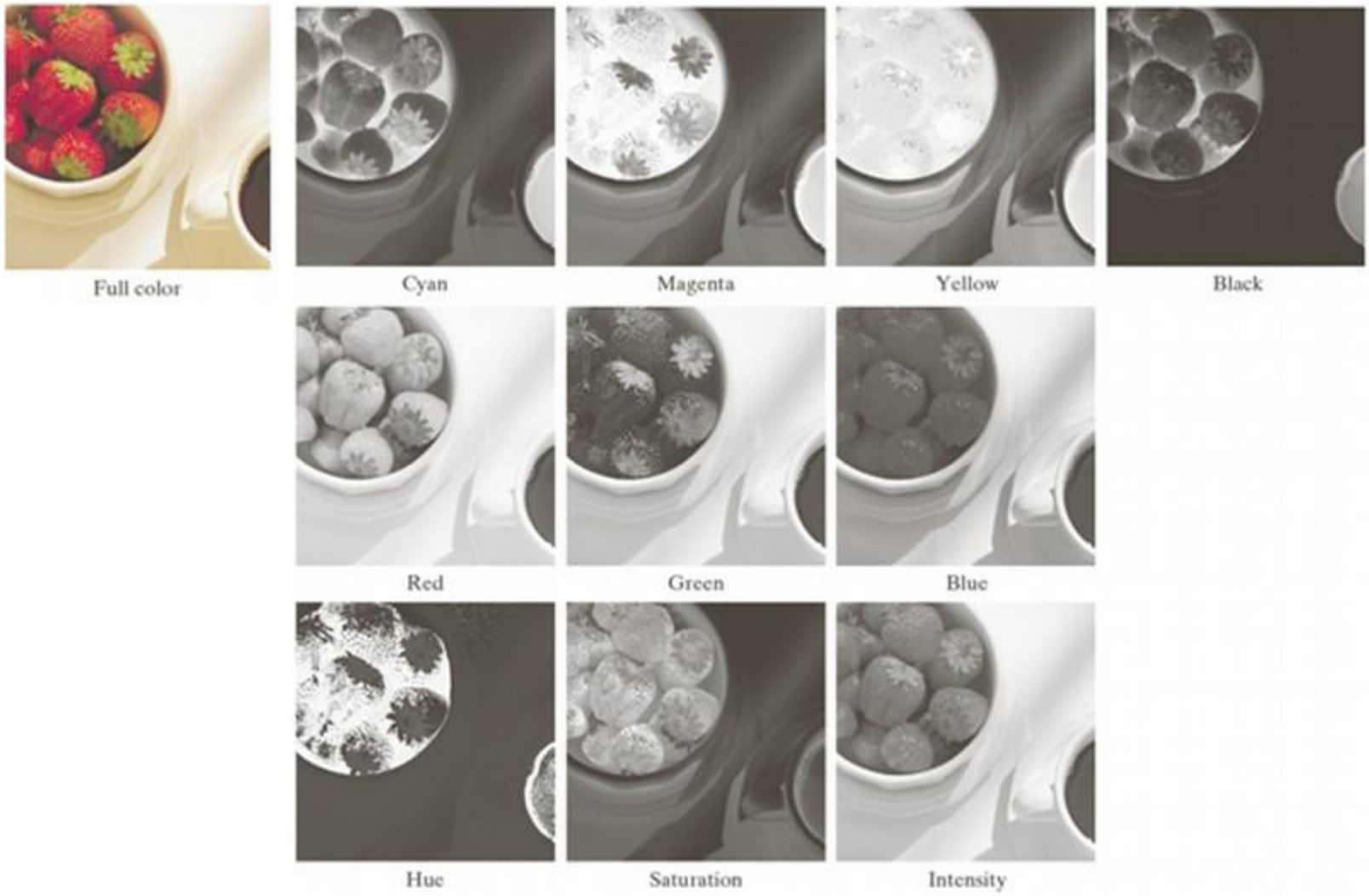
- Full-color image processing consists of two categories:
  - To process each grayscale component individually, then form the composite color image.
  - To process color pixels directly.
- The color pixel can be shown as a vector: $\boldsymbol{c}(x,y)=\begin{bmatrix} c_R(x,y) \\ c_G(x,y) \\ c_B(x,y) \end{bmatrix}=\begin{bmatrix} R(x,y) \\ G(x,y) \\ B(x,y) \end{bmatrix}$



a b

FIGURE 7.27 Spatial neighborhoods for grayscale and RGB color images. Observe in (b) that a *single* pair of spatial coordinates, (x, y), addresses the same spatial location in all three images.

Color Models



Color Transformations

- Color transformation can be modeled as:

$$s_i = T_i(r_i), \quad i = 1, 2, \ldots, n$$

  where **n** is the total number of component images, $\mathbf{r_i}$ are the intensity values of the input component images and $\mathbf{s_i}$ are the spatially corresponding intensities in the output component images and $\mathbf{T_i}$ are the set of ***transformations*** or ***color mappings***.

- **n** is **3** for **RGB**, **CMY** and **HSI**, and is **4** for **CMYK**.
- **T** can be different for each component image.
- Even though any transformation can be performed in any color space, some operations are better suited to specific models.

Color Transformations

- For example, let's modify the intensity of an image with a constant value **k** in the range [0,1].
- In the **HSI** color space, only we should modify the intensity component:

$$\text{HSI:} \quad s_1 = r_1, \quad s_2 = r_2, \quad s_3 = k\, r_3$$

- In **RGB** we modify all three components by the same constant:

$$\text{RGB:} \quad s_i = k\, r_i$$

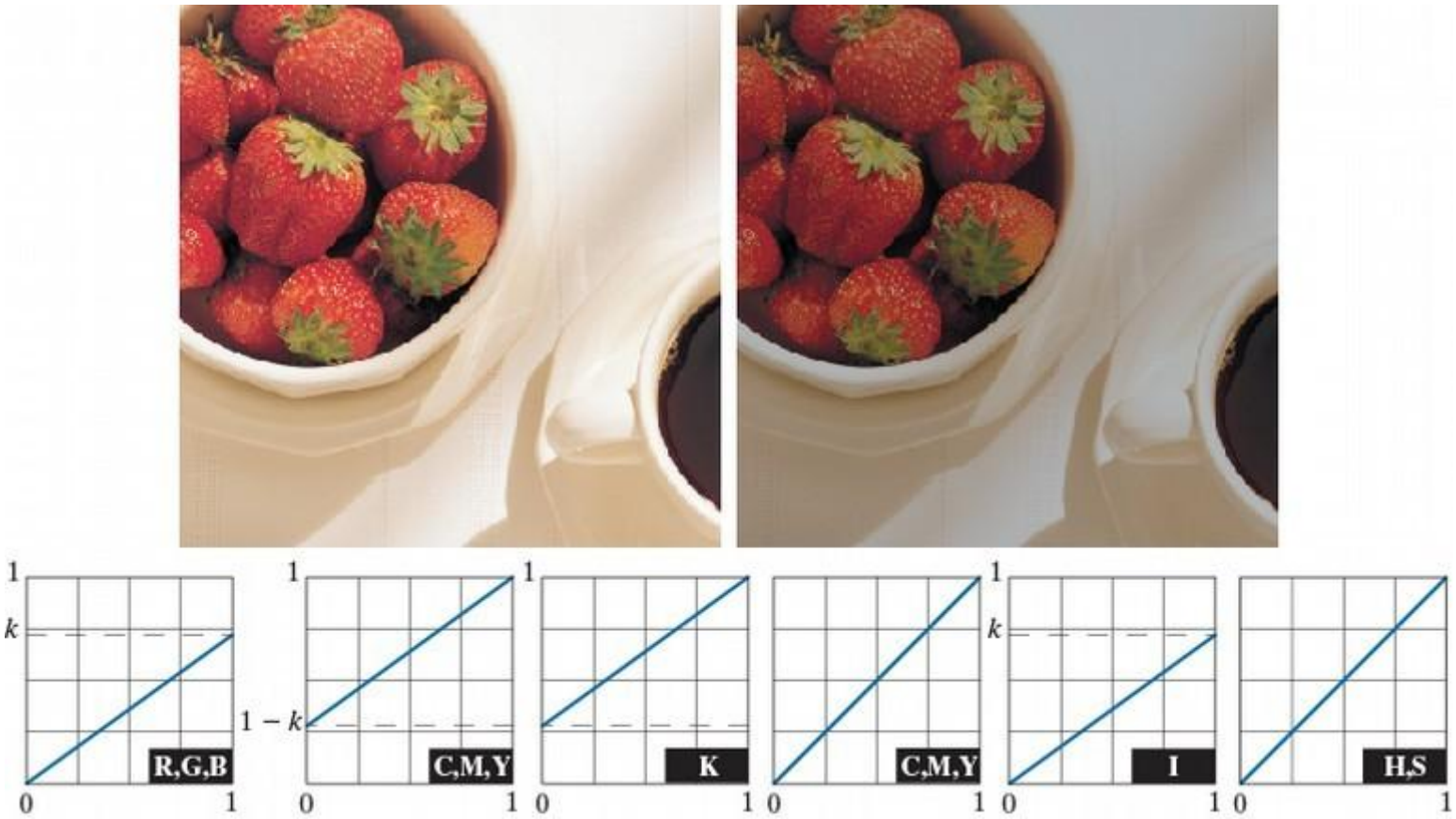
- In **CMY**, all three components are modified similarly as well:

$$\text{CMY:} \quad s_i = k\, r_i + (1-k) \quad i = 1,2,3$$

- In **CMYK**, the transformation is done only by modifying the **K** component:

$$\text{CMYK:} \quad s_i = \begin{cases} r_i & i = 1,2,3 \\ k\, r_i + (1-k) & i = 4 \end{cases}$$

Color Transformations



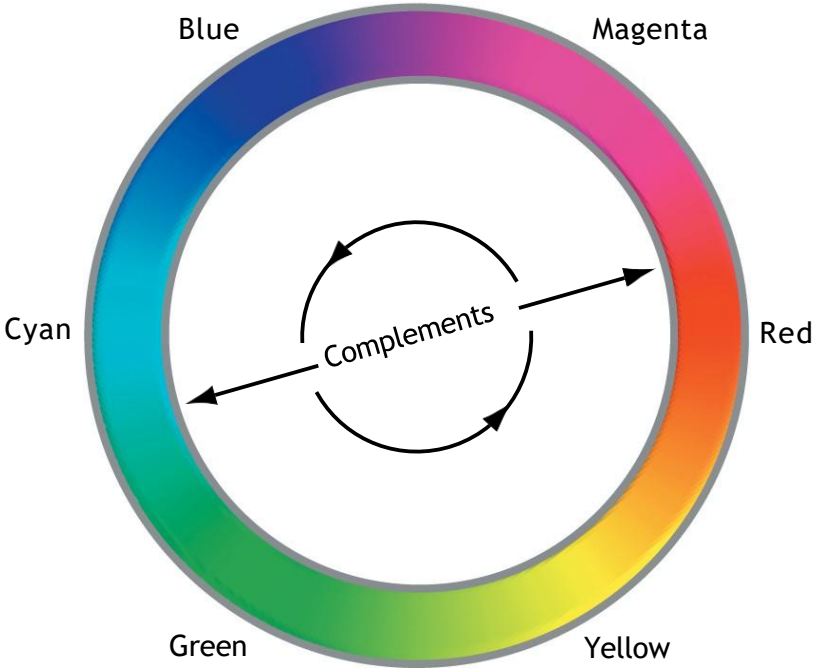
a b
c d e f g h

FIGURE 7.29 Adjusting the intensity of an image using color transformations. (a) Original image. (b) Result of decreasing its intensity by 30% (i.e., letting $k = 0.7$). (c) The required RGB mapping function. (d)–(e) The required CMYK mapping functions. (f) The required CMY mapping function. (g)–(h) The required HSI mapping functions. (Original image courtesy of MedData Interactive.)

Color Complements

- The ***color circle***, or the ***color wheel***, is a visual representation of colors according to their chromatic relationship.
- Hues directly opposite one another on the color circle are complements.
- In **RGB**, the implementation is as:

$$\begin{bmatrix} R_{\text{out}} \\ G_{\text{out}} \\ B_{\text{out}} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R_{\text{in}} \\ G_{\text{in}} \\ B_{\text{in}} \end{bmatrix}$$



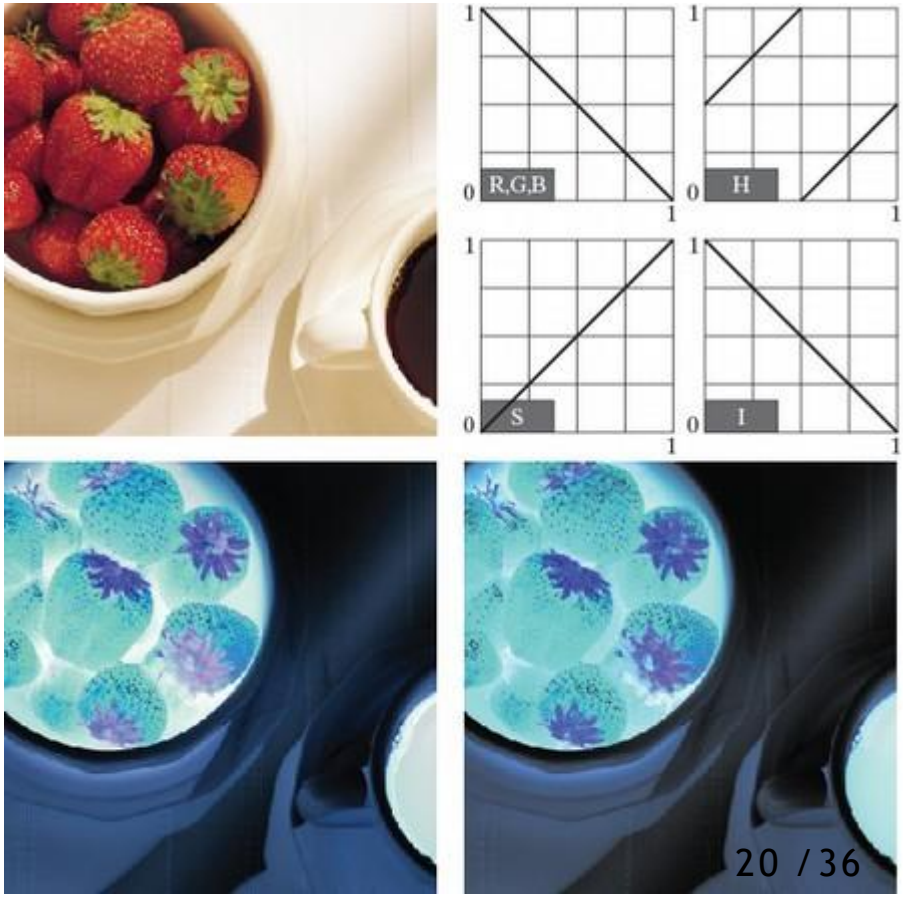
Color Complements

- In **HSI** color space, the complement transformation is not as straightforward.
- The **saturation** component of the **complement** cannot be computed only by the **saturation component** of the input image.
- The implementation is as follows:

$$H_{\text{out}} = \begin{cases} H_{\text{in}} + 0.5 & 0.0 \le H_{\text{in}} \le 0.5 \\ H_{\text{in}} - 0.5 & 0.5 \in H_{\text{in}} \le 1.0 \end{cases}$$

$$S_{\text{out}} = S_{\text{in}}$$

$$I_{\text{out}} = 1 - I_{\text{in}}$$



a b
c d

FIGURE 7.31 Color complement transformations. (a) Original image. (b) Complement transformation functions. (c) Complement of (a) based on the RGB mapping functions. (d) An approximation of the RGB complement using HSI transformations.

Color Slicing

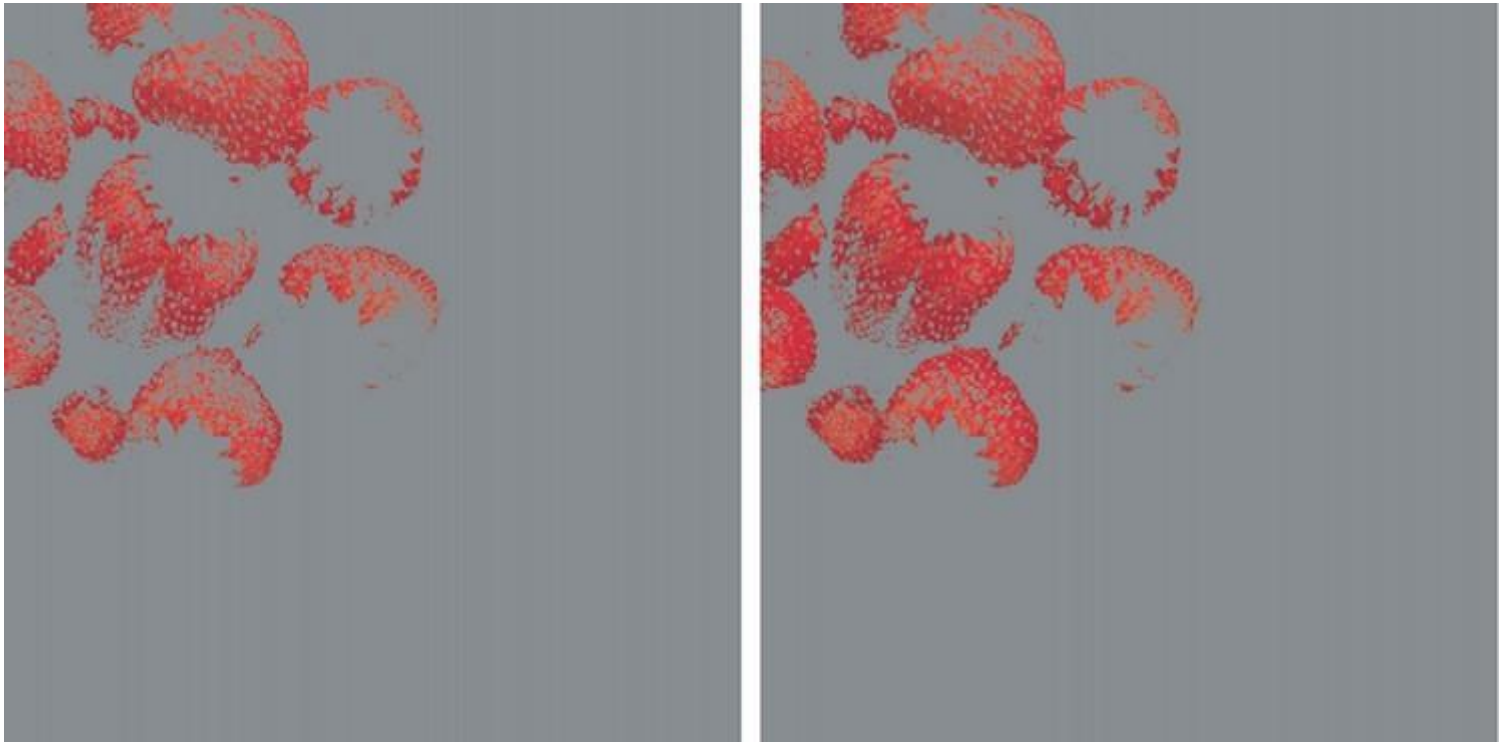
- Highlighting a specific range of colors to display.
- If the colors of interest are shown by a cube (hyper-cube for **n>3**) of width **W**, centered at color **(a_1, a_2, …, a_n)**, the transformations are:

$$s_i = \begin{cases} 0.5 & \text{if } \left[|r_j - a_j| > \frac{W}{2} \right]_{any\, 1 \le j \le n} \\ r_i & otherwise \end{cases} \quad \text{for } i = 1, 2, \ldots, n$$

- If the colors of interest are shown by a sphere (hyper-sphere for **n>3**) of radius **R_0**, centered at color **(a_1, a_2, …, a_n)**, the transformations are:

$$s_i = \begin{cases} 0.5 & \text{if } \sum_{j=1}^{n} (r_j - a_j)^2 > R_0^2 \\ r_i & otherwise \end{cases} \quad \text{for } i = 1, 2, \ldots, n$$

Color Slicing



a b

FIGURE 7.32 Color-slicing transformations that detect (a) reds within an RGB cube of width $W = 0.2549$ centered at $(0.6863, 0.1608, 0.1922)$, and (b) reds within an RGB sphere of radius 0.1765 centered at the same point. Pixels outside the cube and sphere were replaced by color $(0.5, 0.5, 0.5)$.

Tonal and Color Corrections

- Typically the same transformation function is applied to each component if the **RGB** or **CMY** is used.
- For **HSI**, only the **intensity** component is modified while the hue and saturation remain unchanged.

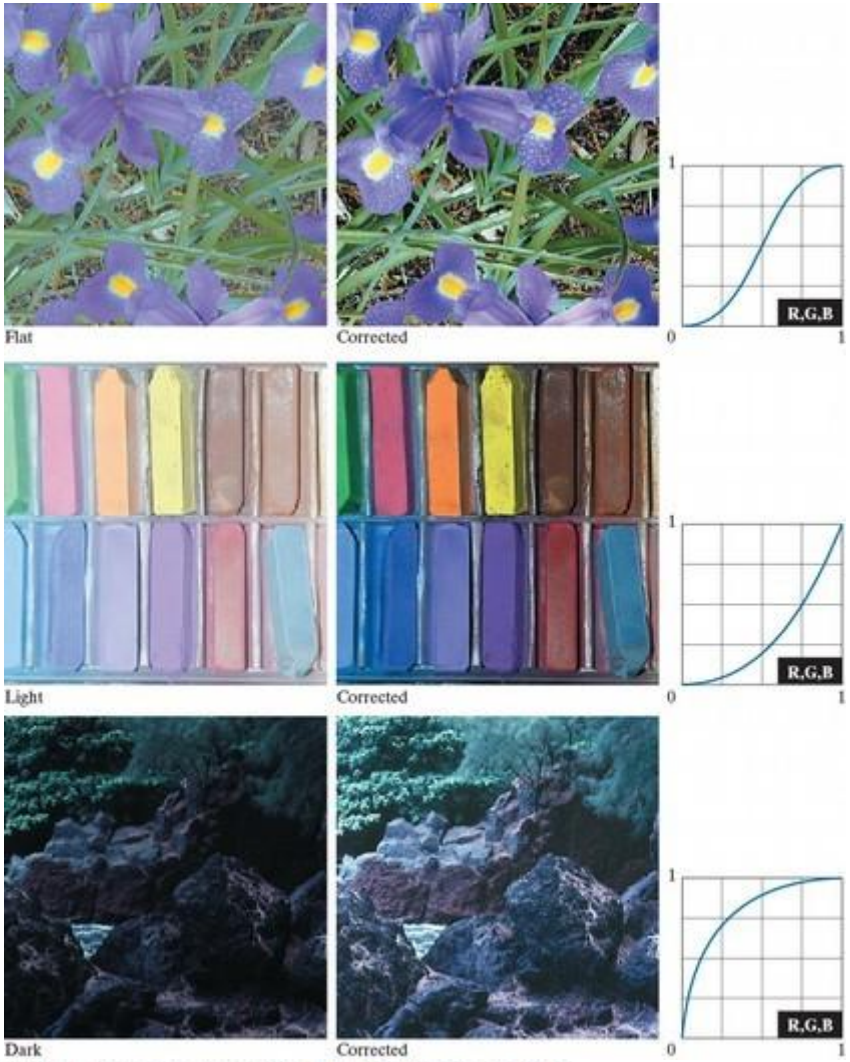


FIGURE 7.33 Tonal corrections for flat, light (high key), and dark (low key) color images. Adjusting the red, green, and blue components equally does not always alter the image hues significantly.

Tonal and Color Corrections

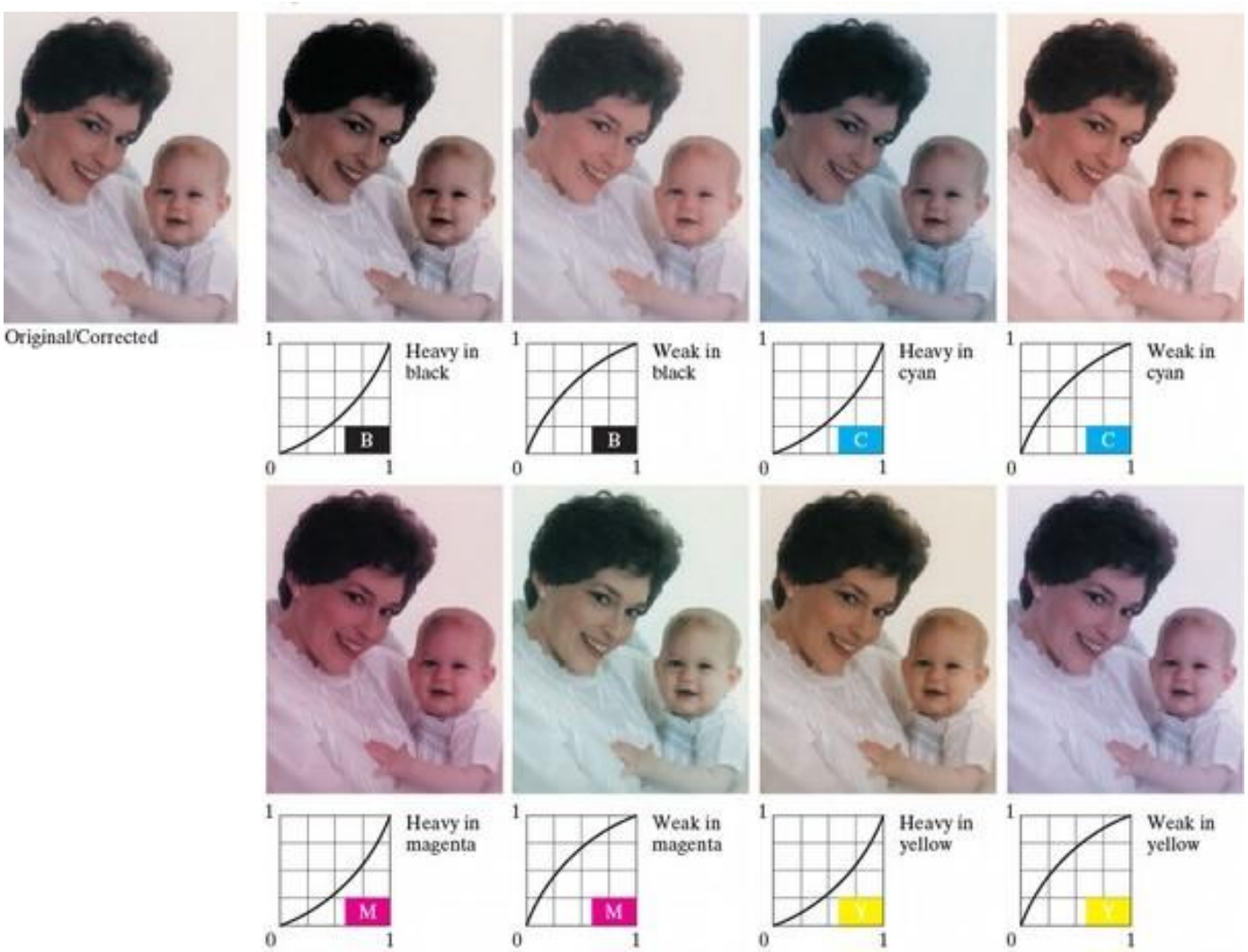
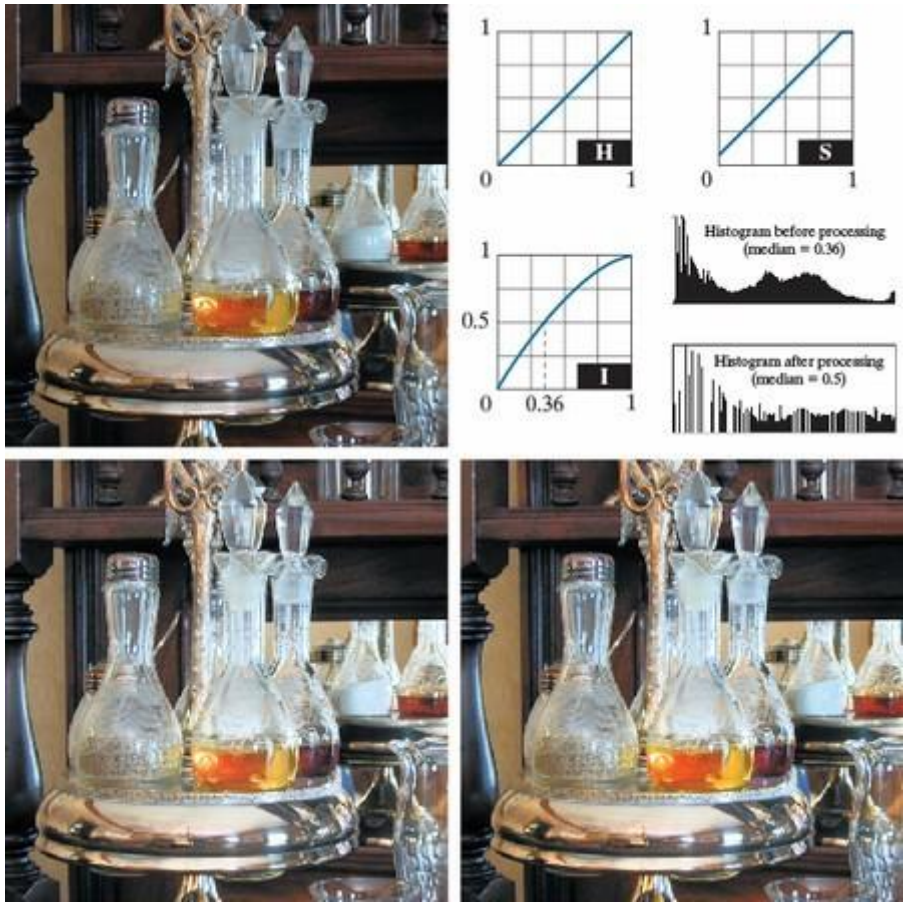


FIGURE 7.34 Color balancing a CMYK image.

Histogram Processing

- Histogram processing of individual color components is not recommended.
- In **HSI**, we can process the intensity component only, and leave the hue and saturation components unchanged.



a b
c d

FIGURE 7.35 Histogram equalization (followed by saturation adjustment) in the HSI color space.

Color Image Smoothing

- In **RGB** color space, smoothing can be done on individual image's color components.
- In **HSI** color space, the intensity and color information are decoupled, so we can only smooth the intensity component.

a b c

FIGURE 7.38 Image smoothing with a 5×5 averaging kernel. (a) Result of processing each RGB component image. (b) Result of processing the intensity component of the HSI image and converting to RGB. (c) Difference between the two results.

Color Image Sharpening

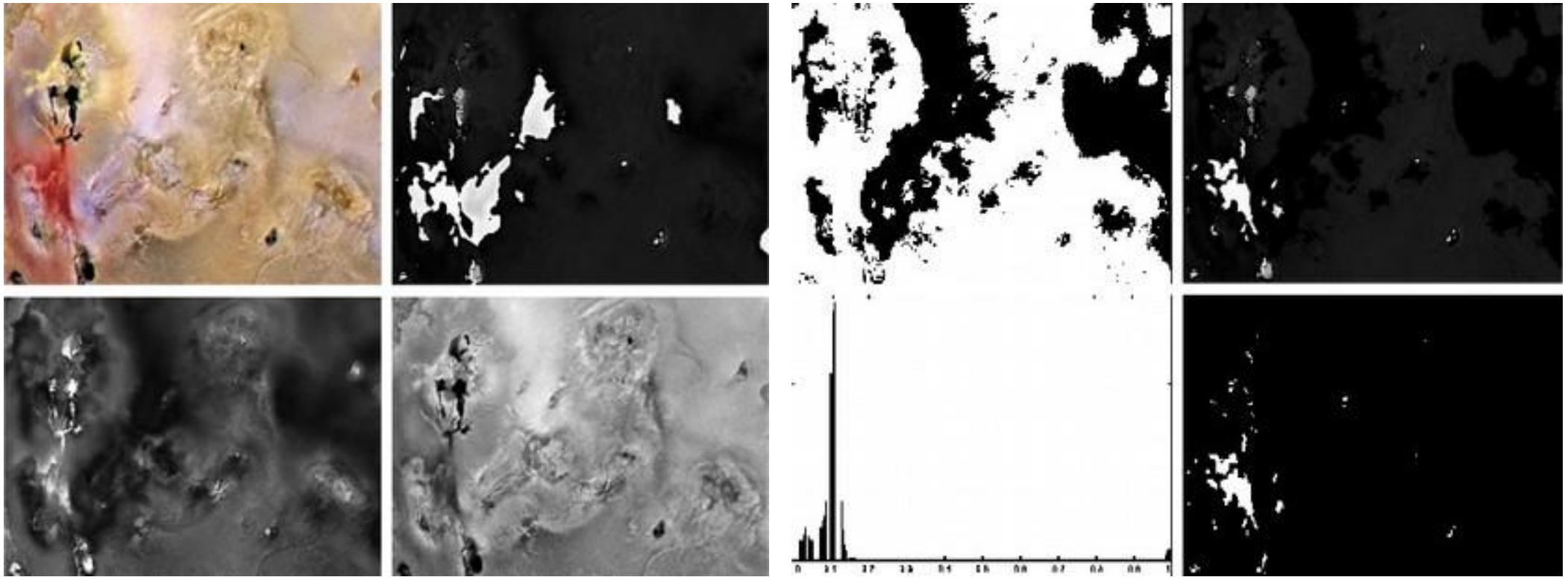
- In **RGB**, the Laplacian operator can be computed for each of the image's color components.
- In **HSI**, only Laplacian of the intensity component is computed for image sharpening.

a b c

FIGURE 7.39 Image sharpening using the Laplacian. (a) Result of processing each RGB channel. (b) Result of processing the HSI intensity component and converting to RGB. (c) Difference between the two results.

Color Image Segmentation

- In **HSI**, saturation is commonly used as a masking image in order to isolate further regions of interest in the hue image.
- The intensity image is less common since it does not carry color information.

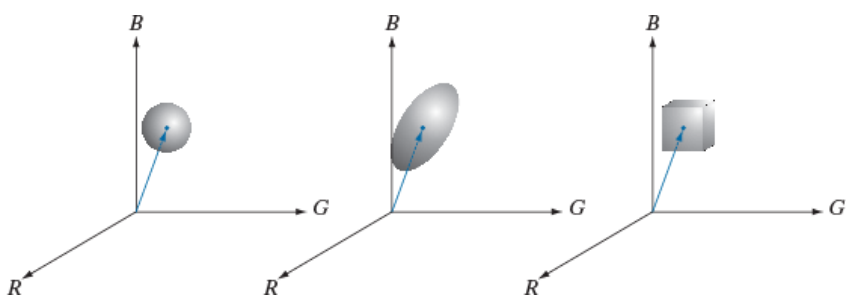


a b e f
c d g h

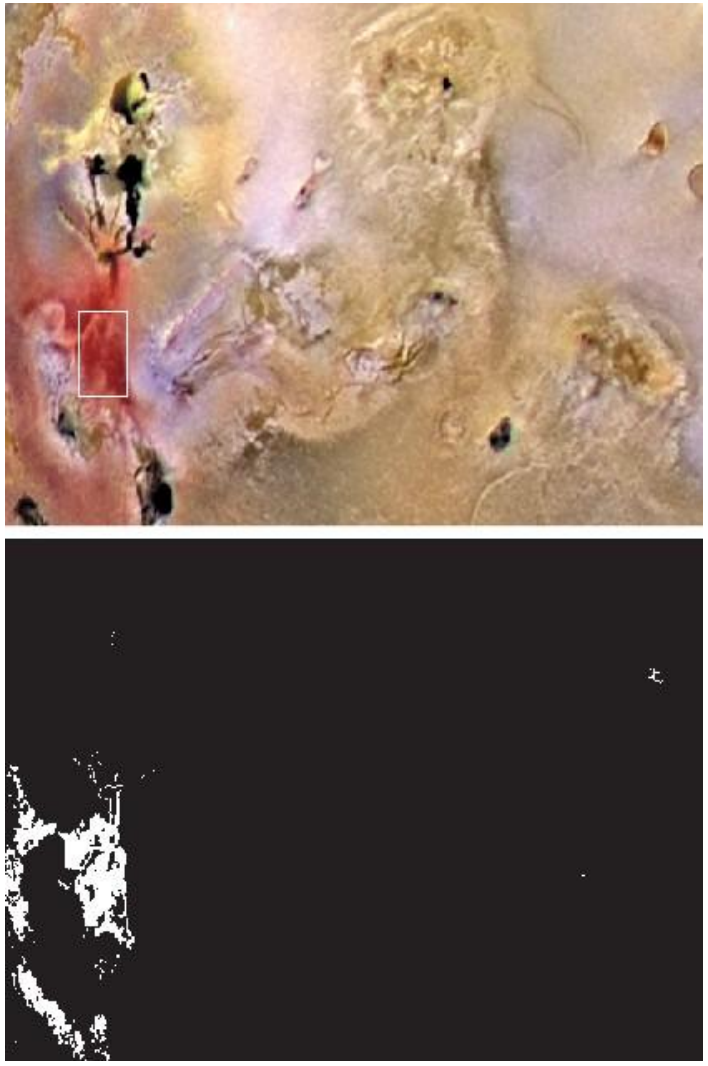
FIGURE 7.40 Image segmentation in HSI space. (a) Original. (b) Hue. (c) Saturation. (d) Intensity. (e) Binary saturation mask (black = 0). (f) Product of (b) and (e). (g) Histogram of (f). (h) Segmentation of red components from (a).

Color Image Segmentation

- Even though working in **HSI** space is more intuitive, segmentation is better if done in **RGB** color space.
- Given a set of sample color points for the color of interest, we can use a distance function to measure the distance of all of the color points to the color of interest, and segment the image by applying a threshold on the distance.



a b c
FIGURE 7.41 Three approaches for enclosing data regions for RGB vector segmentation.



a
b
FIGURE 7.42 Segmentation in RGB space. (a) Original image with colors of interest shown enclosed by a rectangle. (b) Result of segmentation in RGB vector space. Compare with Fig. 7.40(h).

Color Edge Detection

- The gradient discussed before for edge detection of grayscale images cannot be applied directly to color images, since it was not defined for vector quantities.
- Assume **r**, **g** and **b** as the unit vectors along the **R**, **G** and **B** axis of **RGB** color space, and define:

$$\begin{cases} \boldsymbol{u} = \dfrac{\partial R}{\partial x}\boldsymbol{r} + \dfrac{\partial G}{\partial x}\boldsymbol{g} + \dfrac{\partial B}{\partial x}\boldsymbol{b} \\ \boldsymbol{v} = \dfrac{\partial R}{\partial y}\boldsymbol{r} + \dfrac{\partial G}{\partial y}\boldsymbol{g} + \dfrac{\partial B}{\partial y}\boldsymbol{b} \end{cases} \Rightarrow \begin{cases} g_{xx} = \vec{\boldsymbol{u}}.\vec{\boldsymbol{u}} = \boldsymbol{u}^T\boldsymbol{u} = \left|\dfrac{\partial R}{\partial x}\right|^2 + \left|\dfrac{\partial G}{\partial x}\right|^2 + \left|\dfrac{\partial B}{\partial x}\right|^2 \\ g_{yy} = \vec{\boldsymbol{v}}.\vec{\boldsymbol{v}} = \boldsymbol{v}^T\boldsymbol{v} = \left|\dfrac{\partial R}{\partial y}\right|^2 + \left|\dfrac{\partial G}{\partial y}\right|^2 + \left|\dfrac{\partial B}{\partial y}\right|^2 \\ g_{xy} = \vec{\boldsymbol{u}}.\vec{\boldsymbol{v}} = \boldsymbol{u}^T\boldsymbol{v} = \dfrac{\partial R}{\partial x}\dfrac{\partial R}{\partial y} + \dfrac{\partial G}{\partial x}\dfrac{\partial G}{\partial y} + \dfrac{\partial B}{\partial x}\dfrac{\partial B}{\partial y} \end{cases}$$

Color Edge Detection

- Based on the defined quantities, the direction of the **maximum** rate of change of **c(x,y)** is given by:

$$\theta(x,y)=\frac{1}{2}\arctan\left[\frac{2g_{xy}}{g_{xx}-g_{yy}}\right]$$

- The value of the **rate** of change in this direction is:

$$F_{\theta}(x,y)=\left\{\frac{1}{2}\left[(g_{xx}+g_{yy})+(g_{xx}-g_{yy})\cos 2\theta(x,y)+2g_{xy}\sin 2\theta(x,y)\right]\right\}^{\frac{1}{2}}$$

- These equations associate with each point **(x,y)** a pair of orthogonal directions, where along one of them **F** is maximum and it is minimum along the other one.

Color Edge Detection – Example

a b
c d

FIGURE 7.44 (a) RGB image. (b) Gradient computed in RGB color vector space. (c) Gradient image formed by the elementwise sum of three individual gradient images, each computed using the Sobel operators. (d) Difference between (b) and (c).

a b c

FIGURE 7.45 Component gradient images of the color image in Fig. 7.44. (a) Red component, (b) green component, and (c) blue component. These three images were added and scaled to produce the image in Fig. 7.44(c).

Noise in Color Images

- The characteristics of noise are usually similar in all of the image's color components, but this is not always the case.
- Fine grain noise is less visually noticeable in color images than in grayscale images.

a b
c d

FIGURE 7.46 (a)–(c) Red, green, and blue 8-bit component images corrupted by additive Gaussian noise of mean 0 and standard deviation of 28 intensity levels. (d) Resulting RGB image. [Compare (d) with Fig. 7.44(a).]

Noise in Color Images

- Converting an **RGB** image to an **HSI** causes significant degradation of the hue and saturation components, because of the non-linearity of ***cosine*** and ***minimum*** functions in the conversion process.
- The intensity component is slightly smoother, since it is the average of the **RGB** component images.

a b c

FIGURE 7.47 HSI components of the noisy color image in Fig. 7.46(d). (a) Hue. (b) Saturation. (c) Intensity.